

# Generative AI for vaccine misbelief correction: Insights from targeting extraversion and pseudoscientific beliefs

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## ABSTRACT

**Background:** Misinformation about vaccines is a significant barrier to public health, fueling hesitancy and resistance. Generative AI offers a scalable tool for assisting public health communicators in crafting targeted correction messages tailored to audience characteristics. This study investigates the effectiveness of AI-generated messages targeting extraversion and pseudoscientific beliefs compared to high-quality generic and non-vaccine-related messages.

**Method:** In a between-subjects experiment, 1435 U.S. adults were randomly assigned to one of four conditions: control, generic correction, extraversion-targeting correction, or pseudoscientific-belief-targeting correction. Participants rated their agreement with vaccine misbelief statements before and after exposure to a correction message. AI was used to generate the targeted correction messages, while the generic and control messages were sourced from real-world examples.

**Results:** Extraversion-targeting messages significantly reduced vaccine misbeliefs, performing comparably to high-quality generic messages, particularly among participants with higher extraversion levels. However, these effects did not extend to general vaccination attitudes. Pseudoscientific-belief-targeting messages were ineffective and, in some cases, backfired, reinforcing negative attitudes among individuals with strong pseudoscientific beliefs.

**Conclusion:** This study demonstrates the potential of AI-assisted message generation for crafting effective correction messages, particularly when targeting personality traits like extraversion. However, the findings suggest that certain AI-generated messages may be less effective or even counterproductive when targeting entrenched beliefs, underscoring the need for human oversight in refining AI-generated messages. Future research should explore additional audience characteristics and optimize human-AI collaboration to enhance the effectiveness of AI-generated correction messages in public health communication.

## 1. Introduction

Vaccines are among the greatest public health achievements, credited with eradicating smallpox, controlling diseases like polio and measles, and saving millions of lives each year [1]. Despite their proven safety and efficacy, vaccine hesitancy and refusal remain persistent challenges to global health, undermining efforts to control infectious diseases and prevent outbreaks [2]. Individuals exhibiting vaccine hesitancy often delay vaccinations, resist public health policies like mandatory school immunizations, or reject vaccines altogether, leaving themselves and others vulnerable to preventable diseases [3]. Such behaviors threaten community immunity, creating opportunities for once-controlled diseases to reemerge and spread [3].

In today's interconnected digital age, the rapid spread of false claims

online has amplified this challenge [4]. Misinformation and disinformation about vaccines exploit cognitive biases, emotional vulnerabilities, and gaps in scientific literacy, making vaccine-related misbeliefs particularly difficult to counter [5]. While social media platforms have introduced algorithms to curb the spread of misinformation, these efforts have achieved limited success in mitigating its profound influence on vaccine decision-making [5]. Correcting vaccine-related misbeliefs remains an urgent and complex task for public health professionals, requiring innovative strategies to overcome resistance and foster trust in vaccination [5].

Generic correction messages, often grounded in factual rebuttals or scientific evidence, are a common approach but frequently fail to resonate with individuals whose beliefs are deeply rooted in their values, identities, or psychological traits [6]. These misbeliefs often stem from

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mistrust of institutions or emotionally driven narratives that generic messages struggle to address effectively [5,6]. A more promising approach lies in tailoring correction messages to align with the specific characteristics of the audience [6]. By addressing individual needs, values, and cognitive styles, targeted messages can enhance engagement, relevance, and persuasiveness [7]. Despite their potential, targeted messaging strategies for vaccine-related misbeliefs remain mostly underexplored in the literature [6].

Recent advancements in generative AI present a transformative opportunity to implement targeted communication strategies at scale [8,9]. Rather than replacing human expertise, AI serves as an assistive tool that enables public health communicators to efficiently craft and refine personalized correction messages for diverse audiences. Generative AI models, such as large language models (LLMs), can produce highly customized content based on minimal prompts, enhancing the adaptability and reach of misinformation correction efforts [8]. This scalability and flexibility make AI-assisted message generation a compelling tool for tackling the complex challenge of misbelief correction [8,9]. However, the effectiveness of AI-generated targeted messages in addressing deeply entrenched vaccine misbeliefs remains largely unknown.

This study addresses this critical gap by examining the intersection of generative AI and targeted messaging strategies in correcting vaccine-related misbeliefs. Specifically, it investigates the effectiveness of AI-generated correction messages targeting two distinct audience characteristics: extraversion and pseudoscientific beliefs. Extraversion, a personality trait, reflects individual differences in sociability and engagement, potentially shaping how individuals process and respond to information [10]. Pseudoscientific beliefs, on the other hand, represent deeply held misconceptions that are often resistant to correction [11]. By comparing these targeted messages with generic corrections, the study evaluates both the feasibility of AI-assisted targeted communication and the broader potential of tailoring strategies to enhance misbelief correction.

The contributions of this research are both theoretical and practical. Theoretically, it advances understanding of how personalization strategies, such as targeting personality traits or belief systems, can improve the correction of vaccine-related misbeliefs. It also explores the potential and limitations of targeted messaging as a strategy for addressing entrenched misinformation. Practically, the study highlights the role of generative AI as a scalable and flexible tool for public health messaging, offering insights into its strengths and challenges in crafting effective, targeted communication. By addressing these critical gaps, this research paves the way for future innovations in combating misinformation and restoring public trust in vaccines.

### 1.1. Challenges of correcting vaccine misinformation

Efforts to correct misinformation face significant psychological barriers. Once entrenched, misinformation can be challenging to counter, as it often aligns with individuals' existing beliefs, shaping how they perceive and respond to corrective information [12]. Individuals tend to resist contradictory information by generating counterarguments or questioning the credibility of the source [13]. This resistance stems, in part, from the cognitive effort required to update or revise previously held beliefs [12]. While some corrective strategies, such as directly stating that a belief is false, can influence attitudes, they are often insufficient and may, in some cases, backfire by reinforcing the false belief [14]. Although recent studies suggest that backfire effects are less widespread than initially thought, they may still occur among individuals with strongly held beliefs or limited scientific literacy [15].

Despite these challenges, corrections are widely used to address misinformation [16]. Their effectiveness lies in their specificity, as they directly identify and refute the false claim [16]. Corrections can also vary in detail and complexity, depending on the nature of the misinformation being addressed [16]. However, they face two key limitations.

First, corrections are inherently reactive, meaning they address misinformation only after it has already influenced beliefs [16]. Second, their specificity makes them difficult to scale in environments inundated with misinformation [16].

Research on communication interventions countering vaccine misinformation reflects these challenges. A recent systematic review reveals that while some debunking messages effectively reduce belief in misinformation, others show limited or no impact, and in some cases, even reinforce false beliefs [5]. Furthermore, correction-based interventions frequently fail to produce meaningful changes in vaccine attitudes or intentions to vaccinate [5]. The effectiveness of these interventions often varies across different population subgroups, highlighting the role of individual differences. For instance, Moyer-Gusé et al. [17] found that humor-based corrections worked better for individuals with strongly held false beliefs, whereas non-humorous messages were more effective for those already inclined toward positive vaccine attitudes.

This variability underscores the complexity of addressing vaccine hesitancy and emphasizes the need for communication strategies that account for the psychological and demographic diversity of target audiences. Tailoring interventions to align with specific audience characteristics, such as preexisting beliefs or motivational factors, offers a promising alternative to generic corrections [5]. By addressing the underlying psychological roots of vaccine hesitancy, tailored approaches can enhance the salience, relevance, and persuasiveness of interventions [18]. These insights highlight the importance of audience-specific communication strategies and rigorous subgroup analyses in future vaccine communication research.

### 1.2. Potentials of message targeting for correcting vaccine misinformation

Message targeting tailors communication to reflect the characteristics of a specific subgroup, such as their political ideology, values, or identity [19]. It involves purposefully aligning the message content, source, or delivery setting with the audience's features to enhance relevance and resonance [20]. Research on psychological targeting shows that it increases the perceived relevance, familiarity, and fluency of messages, making them more effective in behavior-change interventions [7,20].

Applying message targeting to debunk misbeliefs presents unique challenges due to the cognitive and motivational barriers associated with belief revision. Misbeliefs tied to personal motivations or identity are particularly resistant to correction, as contradictory information can trigger defensiveness or skepticism [12]. However, scholars argue that targeting corrections to align with the audience's values or worldviews offers a more promising alternative to mass communication strategies [18]. Value-congruent framing, where messages resonate with the audience's moral or ideological foundations, has been identified as a particularly effective strategy for reducing resistance and fostering belief revision [12,21].

Initial evidence supports the potential of message targeting in addressing vaccine misinformation. For example, unvaccinated individuals expressed greater trust in medical professionals who first emphasized shared religiosity before discussing vaccines [22]. Similarly, messages affirming individuals' underlying concerns, such as safety or autonomy, fostered greater vaccine trust [23]. Campaigns engaging with alternative lifestyle communities by framing vaccination as congruent with their values reported positive shifts in attitudes [24]. Furthermore, interventions appealing to traits like moral purity or needle aversion, factors strongly associated with vaccine misinformation support, significantly reduced endorsement of false beliefs [6]. These findings suggest that targeting appeals to audiences' personal motivations and identities, thereby increasing their openness to correcting misinformed beliefs.

While these examples highlight the promise of targeting, questions remain about its scalability and broader applicability. Crafting targeted

messages for diverse audiences can be resource-intensive, particularly in environments saturated with misinformation. Advances in generative AI offer a potential solution by enabling the rapid creation of targeted messages at scale. The next section explores how AI technologies can operationalize message targeting, bridging the gap between its potential and the practical demands of large-scale public health interventions.

### 1.3. Generative AI for message targeting at scale

The advent of LLMs has transformed the landscape of message targeting by enabling automated and scalable customization [25]. Traditionally, crafting targeted messages was a labor-intensive process, limiting the feasibility of deploying such strategies on a large scale [26]. Today, LLMs streamline this process, generating personalized messages rapidly and with minimal input. Research also underscores the effectiveness of generative AI in creating targeted messages that outperform generic ones. For instance, ChatGPT has demonstrated consistent success in producing persuasive, targeted communication across various domains, even when provided with brief prompts [8,9]. This capability highlights the potential of AI as an assistive tool for creating impactful messages tailored to specific audience characteristics at scale. Additionally, generative AI offers a unique advantage in efficiency: it can generate multiple variations of a message from a single template, addressing diverse audience traits without requiring extensive manual effort [8]. Advances in machine learning further enhance this process, enabling rapid validation of AI-generated messages by inferring audience traits, such as personality, from their interactions with text [25].

While much progress has been made in leveraging generative AI for persuasion, its application to correcting vaccine misbeliefs remains underexplored. Promising insights come from related fields. For example, a recent study used GPT-4 Turbo to engage over 2000 conspiracy believers in personalized evidence-based dialogues [27]. This intervention reduced conspiracy beliefs by 20 %, with effects persisting two months later and generalizing across multiple conspiracy theories. Importantly, the results extended to participants with deeply entrenched beliefs, showcasing the potential of AI-assisted tailored communication. However, this success relied on extensive, interactive exchanges, which may not be scalable for addressing widespread vaccine misbeliefs.

This raises a critical question: can generative AI effectively address vaccine misbeliefs through simpler interventions, such as a single, targeted correction message? If so, it could unlock new possibilities for combating misinformation and improving public health communication at scale. The current study seeks to address this question, exploring whether AI-generated targeted correction messages can produce meaningful belief changes, thereby advancing the potential of generative AI as a tool for scalable, effective misinformation correction.

## 2. The Present Study

To address the research objectives outlined above, this study employs an experimental design to evaluate the effectiveness of AI-assisted correction messages targeting two distinct audience characteristics: extraversion and pseudoscientific beliefs. Specifically, it compares the effectiveness of three types of correction messages: (1) an extraversion-targeting message generated by AI, (2) a pseudoscientific-belief-targeting message generated by AI, and (3) a generic correction message sourced from a real-world example. These three conditions are further compared to a control condition featuring a non-vaccine-related correction message also drawn from a real-world example. This design allows for an assessment of the impact of targeted and generic corrections on both the general U.S. adult population and subgroups with varying levels of extraversion and pseudoscientific beliefs.

The study focuses on extraversion and pseudoscientific beliefs as the audience characteristics for targeted messages based on their theoretical and practical relevance. Extraversion, a relatively stable personality trait, is included because previous research consistently demonstrates

that messages targeting personality traits are more effective than generic messages in various contexts [7,8]. Moreover, extraversion has been shown to yield stronger effects in targeted messaging compared to other traits such as agreeableness and conscientiousness [8]. This study leverages these findings to test whether targeting a personality trait unrelated to the vaccine context can improve correction effectiveness. In contrast, pseudoscientific beliefs are a belief-based cognition more closely tied to vaccine attitudes and misinformation. These beliefs have been identified as key drivers of anti-vaccination attitudes [11,28,29], making them a critical target for interventions. Addressing pseudoscientific beliefs directly through tailored corrections has the potential to reduce misbeliefs and shift broader vaccine attitudes [28,29]. However, the effectiveness of such an approach remains untested, making this study an important contribution to the literature.

By employing a targeted message approach, this study investigates whether the relevance of the targeted audience characteristics to the correction context influences message effectiveness. Specifically, it compares the impact of a message targeting a personality trait unrelated to vaccine correction (extraversion) with a message targeting a belief closely tied to vaccine misbeliefs (pseudoscientific beliefs). Correction message effectiveness is evaluated using two outcome variables: vaccine misbeliefs and general vaccination attitudes. These outcomes are selected to capture both the immediate and downstream effects of the intervention. Previous research has shown that correction messages tend to have a stronger impact on immediate outcomes, such as reducing vaccine misbeliefs, compared to more distal outcomes like improving general vaccination attitudes [5,16]. Therefore, this study also seeks to explore whether AI-assisted targeted correction messages can influence both types of outcomes.

Based on the literature review and theoretical rationale, the following hypotheses are proposed:

**H1:** The generic condition, extraversion-targeting condition, and pseudoscientific-belief-targeting condition will be more effective than the control condition in (a) reducing vaccine misbeliefs and (b) improving attitudes toward vaccination.

**H2:** The experimental conditions will interact with extraversion such that for individuals high in extraversion, the extraversion-targeting condition will be more effective than the other conditions in (a) reducing vaccine misbeliefs and (b) improving attitudes toward vaccination.

**H3:** The experimental conditions will interact with pseudoscientific beliefs such that for individuals high in pseudoscientific beliefs, the pseudoscientific-belief-targeting condition will be more effective than the other conditions in (a) reducing vaccine misbeliefs and (b) improving attitudes toward vaccination.

## 3. Method

### 3.1. Procedure

This study was conducted following approval from the Institutional Review Boards at the University of Michigan. Participants provided informed consent before beginning the study. They were first presented with a list of statements about vaccination, climate change, and genetically modified organisms (GMOs), all of which reflected common misbeliefs. Participants were asked to rate their agreement with each statement. Among these, three statements about vaccines (described in detail under the “Measures” section) were included because they represent pseudoscientific beliefs and had received high endorsement and familiarity ratings in prior studies with U.K. and U.S. samples [28,29]. Statements on climate change and GMOs were included to obscure the true focus of the study and reduce potential demand characteristics [30].

Participants then completed measures assessing their pseudoscientific beliefs, extraversion, and other cognitions and personality traits. Afterward, they were randomly assigned to one of four experimental

conditions (control, generic, extraversion-targeting, or pseudoscientific-belief-targeting) as part of a one-factor, four-level between-subjects factorial design. Participants were told they would read a misinformation-correction message from a mainstream fact-checking website.

In the control condition, participants read a message refuting a false claim unrelated to vaccines or health (see “3.3 Design and stimuli” for details). In the generic, extraversion-targeting, and pseudoscientific-belief-targeting conditions, participants received a correction message tailored to the specific nature of the assigned condition. For these three conditions, the message addressed the participant's strongest vaccine misbelief, defined as the misbelief statement receiving the highest endorsement rating at the beginning of this study [23]. If multiple misbeliefs tied for the highest endorsement, one was randomly selected.

After reading the assigned correction message, participants re-evaluated their agreement with the same list of statements about vaccination, climate change, and GMOs. Additional survey questions assessed vaccination attitudes and other key measures described below. At the conclusion of the experiment, participants were debriefed on the study's true purpose and design. Fig. 1 depicts the experimental design and procedure.

### 3.2. Participants

In November 2024, a national sample of U.S. adults ( $N = 1452$ ) was recruited via Prolific and compensated \$1.80 for their participation in this study, which had an average duration of 11 min. To ensure that participants would endorse at least one vaccine misbelief tested in the study, prescreening was used to include only individuals who felt neutral or negative toward COVID-19 vaccines [28]. Participants who failed at least one of two attention-check questions were excluded, resulting in a final sample of 1435 participants. An *a priori* power analysis conducted in G\*Power for a one-factor, four-level between-subjects factorial design with a small effect size ( $f = 0.10$ ),  $\alpha = 0.05$ , and 0.80 power indicated that 1096 participants were sufficient to detect main effects, making the final sample size of 1435 ideal for this study.

Of the final sample, 37.2 % identified as male, 61.4 % as female, and 1.4 % as nonbinary or preferred not to answer. Participants had an average age of 40.8 years ( $SD = 12.5$ ), a median education level of an associate degree, and a median household income between \$50,000 and \$74,999 annually. The racial/ethnic breakdown was as follows: 64.9 % White, 17.3 % Black, 7.1 % Hispanic or Latine, 5.7 % Asian, 1.0 % American Indian or Alaska Native, 0.3 % Native Hawaiian or Pacific Islander, 3.0 % other, and 0.6 % preferred not to answer. Notably, 43.6 % of participants reported never receiving a COVID-19 vaccine.

### 3.3. Design and stimuli

All correction messages followed a standardized claim-explanation-verdict format, modeled after formats commonly used on fact-checking websites like [fullfact.org](https://fullfact.org) and [reuters.com/fact-check/](https://reuters.com/fact-check/). Messages ranged in length from 150 to 220 words, depending on the condition and the misbelief being addressed (see Fig. 2 for an example and see Appendix A for full messages).

The control message was adapted from a Reuters Fact Check article refuting a false claim that migrants arriving in small boats in England receive free access to historic sites. This message was selected because it was unrelated to vaccines or health and was psychologically distant from participants in terms of both geography and topic.

The generic correction messages were adapted from [jitsuvax.info](https://jitsuvax.info), a website dedicated to countering vaccine misinformation through empirically supported refutations [18,28,29]. Messages from this source were selected because they are accessible to the general public and therefore suitable as high-quality, non-targeted corrections, further serving as a relatively conservative test of the effectiveness of the more

targeted correction messages. Each generic message was reformatted to match the structure of the other conditions.

Correction messages for the extraversion-targeting and pseudoscientific-belief-targeting conditions were generated using the “temporary chat” function of GPT-4o to ensure no prior usage influenced the generated content. For extraversion-targeting messages, adapted from Simchon, Edwards [9], the prompt was: “One way in which people differ from each other is through their personalities. The Big-Five Personality Model identifies five personality traits on which each individual varies. One of the traits is Extraversion. People high in Extraversion tend to be sociable, talkative, assertive, and full of energy. Please rephrase the following message from a fact-checking website so it would appeal to people high in Extraversion. Keep the basic format the same.”

For pseudoscientific-belief-targeting messages, adopting the definition from Fasce, Avendaño [11], the prompt was: “Pseudoscience is usually defined as cognitions involving radical epistemic misconduct and science mimicry, with two major forms: pseudo-theory promotion and science denial. On one hand, pseudo-theory promotion is based on complicated doctrinal content, which often includes flawed mechanistic explanations and insubstantial conceptual schemes. On the other hand, science deniers tend to concoct false controversies, motivated by their enmity towards a well-established scientific theory. Please rephrase the following message from a fact-checking website so it would appeal to people with strong beliefs in pseudoscience. Keep the basic format the same.” This process was repeated for each of the three vaccine misbeliefs.

### 3.4. Measures<sup>1</sup>

**Vaccine Misbeliefs.** Participants rated their agreement with three vaccine-related statements (1 = *Strongly disagree*, 7 = *Strongly agree*) both before and after reading the correction message. These statements [28] were: “Vaccines interfere with the body's natural immunity, which is a better defense against the disease” (before:  $M = 4.05$ ,  $SD = 1.80$ ; after:  $M = 3.84$ ,  $SD = 1.87$ ); “People are being offered too many vaccines nowadays, and this will overload their immune systems” (before:  $M = 4.45$ ,  $SD = 1.80$ ; after:  $M = 4.06$ ,  $SD = 1.91$ ); “Instead of vaccines, people should improve environmental factors like good hygiene, healthy lifestyles, and protective measures against the disease” (before:  $M = 4.64$ ,  $SD = 1.70$ ; after:  $M = 4.23$ ,  $SD = 1.87$ ).

**Changes in Vaccine Misbeliefs.** For each condition, a target misbelief was identified. In the generic, extraversion-targeting, and pseudoscientific-belief-targeting conditions, the target misbelief corresponded to the specific belief refuted in the correction message. In the control condition, where no vaccine-related correction was provided, the target misbelief was randomly selected if participants exhibited a tie for the strongest endorsement across the three vaccine-related misbeliefs. To measure changes in misbeliefs, a difference score ( $M = -0.91$ ,  $SD = 1.29$ ) was calculated by subtracting the pre-exposure agreement rating from the post-exposure agreement rating. A negative difference score indicated a reduction in the targeted misbelief, while a positive score reflected an increase.

**Vaccination Attitude.** General attitudes toward vaccination were assessed using three items: “How positive or negative/favorable or unfavorable/supportive or unsupportive do you feel about vaccination in general?” (1 = *Very negative/unfavorable/unsupportive*, 5 = *Very positive/*

<sup>1</sup> This study also assessed the perceived credibility of the correction messages using items such as accuracy, trustworthiness, and believability. A one-way ANOVA revealed no significant differences in perceived credibility among the four conditions,  $F(3, 1429) = 1.524$ ,  $p = .206$ , partial  $\eta^2 = 0.003$ . These results suggest that participants did not systematically perceive the AI-generated messages (i.e., extraversion-targeting and pseudoscientific-belief-targeting messages) as less credible than the real-world fact-checking messages (i.e., control and generic messages).

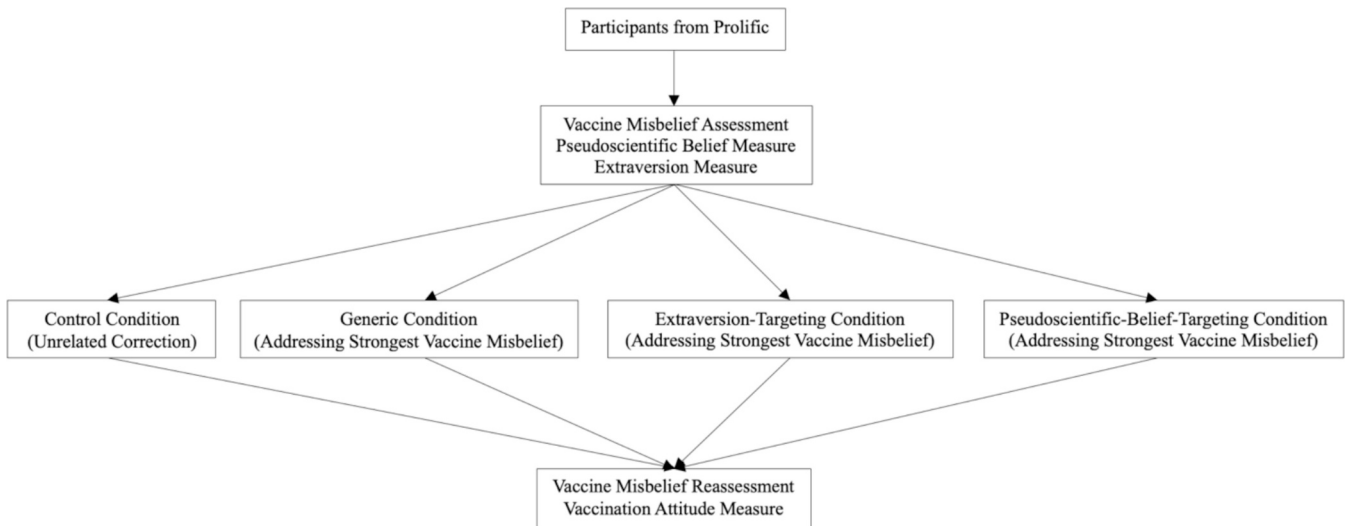


Fig. 1. Experimental Design and Procedure.

**Claim:** Vaccines interfere with the body’s natural immunity, which is a better defense against disease.

**Fact:** This claim lacks any credible evidence.

**Explanation:** Vaccines are like giving your immune system a power boost! They work with your body’s natural defenses, making them stronger and more prepared for serious diseases that could otherwise knock you off your feet. Think of it like giving your immune system a heads-up before the fight even starts, keeping you energized and ready to tackle whatever comes your way.

For instance, COVID-19 vaccines have saved *almost half a million lives* in under a year by helping people avoid the worst symptoms. That means you get natural immunity the smart way—without going through the exhausting and dangerous illness. Vaccines are your immune system’s best workout buddy, getting you pumped up to take on anything!

**Verdict:** False. There’s no solid evidence that vaccines interfere with natural immunity or that your natural immunity alone is a better defense. Getting vaccinated is simply the smarter, safer way to stay strong and healthy!

Fig. 2. An Extraversion-Targeting Message Generated by AI.

favorable/supportive;  $\alpha = 0.96$ ;  $M = 3.25$ ,  $SD = 1.18$ ).

**Extraversion.** Participants rated their agreement (1 = *Strongly disagree*, 5 = *Strongly agree*) with items measuring extraversion traits [10]. Sample items included: “I am outgoing and sociable” and “I am full of energy” ( $\alpha = 0.77$ ;  $M = 3.06$ ,  $SD = 0.83$ ).

**Pseudoscientific Beliefs.** Participants rated their agreement (1 = *Strongly disagree*, 5 = *Strongly agree*) with pseudoscientific belief items [11], such as “All the cells of our body store memories (cellular memories), ours or of our ancestors” ( $\alpha = 0.80$ ;  $M = 2.98$ ,  $SD = 0.98$ ).

### 3.5. Analyses

For H1, one-way ANOVAs were conducted with the four experimental conditions as the independent variable and changes in vaccine misbeliefs and vaccination attitudes as the dependent variables. LSD post hoc analyses were used for pairwise comparisons following significant main effects.

Regarding H2 and H3, moderation analyses were performed using Model 1 in the PROCESS macro for SPSS [31]. For H2, the independent variables included the experimental conditions (as a multi-categorical variable in the analysis), the extraversion scale, and their interaction,

with the extraversion-targeting condition as the reference group. For H3, the pseudoscientific-belief scale replaced the extraversion scale, and the pseudoscientific-belief-targeting condition served as the reference group. Changes in vaccine misbeliefs and vaccination attitudes served as the dependent variables in both cases. The Johnson-Neyman method was used to identify regions of significance.

## 4. Results

### 4.1. Main effects of experimental conditions (H1)

The ANOVA results revealed a significant main effect of the experimental conditions on changes in vaccine misbeliefs,  $F(3, 1431) = 38.70$ ,  $p < .001$ , partial  $\eta^2 = 0.075$ . Participants in the generic condition ( $M = -1.19$ ,  $SD = 1.49$ ), the extraversion-targeting condition ( $M = -1.20$ ,  $SD = 1.36$ ), and the pseudoscientific-belief-targeting condition ( $M = -0.95$ ,  $SD = 1.24$ ) showed significantly greater reductions in vaccine misbeliefs compared to the control condition ( $M = -0.35$ ,  $SD = 0.84$ ),  $ps < 0.001$ . These findings support **H1a**, demonstrating the overall effectiveness of vaccine-related correction messages compared to a non-vaccine-related control message.

Unexpectedly, both the generic condition and the extraversion-targeting condition were significantly more effective in reducing vaccine misbeliefs than the pseudoscientific-belief-targeting condition ( $p_s = 0.009$ ), indicating that targeting pseudoscientific beliefs may not offer additional benefits over more general or personality-targeted messages. However, no significant effects of the experimental condition emerged for general attitudes toward vaccination,  $F(3, 1431) = 0.19$ ,  $p = .904$ , partial  $\eta^2 = 0.000$ . Thus, **H1b** was not supported.

#### 4.2. Moderation by extraversion (H2)

For changes in vaccine misbeliefs, the moderation analysis revealed significant two-way interactions between extraversion and the control condition ( $B = 0.30$ ,  $SE = 0.11$ ,  $p = .005$ ), between extraversion and the generic condition ( $B = 0.25$ ,  $SE = 0.11$ ,  $p = .026$ ), and between extraversion and the pseudoscientific-belief-targeting condition ( $B = 0.27$ ,  $SE = 0.12$ ,  $p = .021$ ). Fig. 3 presents these moderation results.

For the comparison between the control and extraversion-targeting conditions, the extraversion-targeting condition became significantly more effective in correcting vaccine misbeliefs when participants' extraversion scores exceeded 1.53, representing individuals with relatively low extraversion. Similarly, when comparing the pseudoscientific-belief-targeting condition with the extraversion-targeting condition, the extraversion-targeting condition outperformed the pseudoscientific-belief-targeting condition among individuals with extraversion scores higher than 2.89, representing those with moderate levels of extraversion. A more complex pattern emerged when comparing the generic and extraversion-targeting conditions. For participants with extraversion scores below 1.57, the generic condition was more effective in correcting vaccine misbeliefs than the extraversion-targeting condition. However, for participants with extraversion scores above 4.70, the extraversion-targeting condition was more effective than the generic condition.

These findings provide support for **H2a**, demonstrating the superiority of the extraversion-targeting condition for individuals with high levels of extraversion. However, no significant interaction effects emerged for general attitudes toward vaccination, providing no support for **H2b**.

#### 4.3. Moderation by pseudoscientific beliefs (H3)

The analysis revealed no significant interaction effects between pseudoscientific beliefs and experimental conditions for changes in vaccine misbeliefs, indicating that pseudoscientific beliefs did not

moderate the effectiveness of the correction messages in this context. Consequently, **H3a** was not supported.

For general attitudes toward vaccination, a significant two-way interaction was found between pseudoscientific beliefs and the control condition ( $B = 0.22$ ,  $SE = 0.09$ ,  $p = .010$ ). Interestingly, among participants with pseudoscientific belief scores below 2.37, representing those with moderate-to-low pseudoscientific beliefs, the pseudoscientific-belief-targeting condition led to more positive vaccination attitudes than the control condition. However, for participants with pseudoscientific belief scores above 4.87, representing those with strong pseudoscientific beliefs, the pseudoscientific-belief-targeting condition backfired, resulting in more negative vaccination attitudes compared to the control condition. These results suggest that targeting pseudoscientific beliefs may inadvertently reinforce negative attitudes among individuals with entrenched pseudoscientific beliefs. Thus, no support was found for **H3b**.

#### 4.4. Post-hoc human validation

To further assess whether the AI-generated correction messages effectively reflected qualities that would appeal to individuals high in extraversion or with strong beliefs in pseudoscience, post-hoc human validation studies were conducted. Following the validation procedure outlined by Simchon, Edwards [9], two separate validation studies were carried out with one for extraversion-targeting messages and one for pseudoscientific-belief-targeting messages. Each study recruited 50 U.S. adults via Prolific.

Participants were shown three correction messages, each addressing one of the three vaccine misbeliefs examined in this study. For each message, participants were randomly assigned to view either the generic version or the AI-generated targeted version. After reading the message, participants rated the extent to which they believed it would appeal to individuals high in extraversion (for the extraversion-targeting validation study) or to those with strong beliefs in pseudoscience (for the pseudoscientific-belief-targeting validation study). Ratings were provided on a seven-point scale (1 = *Not at all*, 7 = *Extremely*). To ensure consistency with the AI's message-generation process, the same descriptions of extraversion and pseudoscientific beliefs used as prompts for AI-generated messages were also provided to participants in the validation studies.

Results from one-way ANOVAs supported the effectiveness of the targeted messages. The extraversion-targeting messages ( $M = 4.84$ ,  $SD = 1.59$ ) were perceived as more appealing to people high in extraversion than the generic messages ( $M = 3.92$ ,  $SD = 1.70$ ),  $F(1, 148) = 11.67$ ,  $p <$

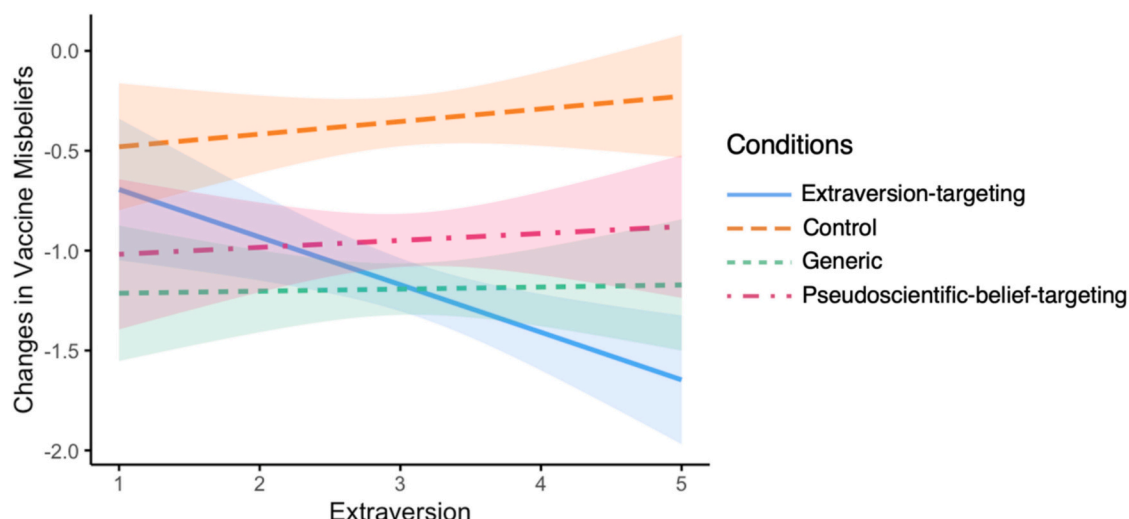


Fig. 3. Interaction Effects Between Experimental Conditions and Extraversion.

.001, partial  $\eta^2 = 0.073$ . Similarly, the pseudoscientific-belief-targeting messages ( $M = 4.23$ ,  $SD = 1.92$ ) were rated as more appealing to people with strong pseudoscientific beliefs compared to the generic messages ( $M = 3.56$ ,  $SD = 1.86$ ),  $F(1, 148) = 4.581$ ,  $p = .034$ , partial  $\eta^2 = 0.030$ .

These findings validate that the AI-generated messages successfully incorporated the intended characteristics, reinforcing their suitability for targeting specific audience traits.

## 5. Discussions

This study provides valuable insights into the effectiveness of AI-assisted correction messages tailored to audience characteristics, contributing to the growing body of research on misinformation correction and personalized communication. By examining the effects of messages targeting extraversion and pseudoscientific beliefs, this study advances our understanding of the potential and challenges of integrating generative AI as a tool to enhance public health communication. The findings offer theoretical contributions, practical implications, and directions for future research.

### 5.1. Effectiveness of extraversion-targeting messages

The results for the extraversion-targeting condition highlight the promise of tailoring messages to audience personality traits. AI-generated messages that targeted extraversion were as effective as, and in some cases more effective than, high-quality generic correction messages in reducing vaccine misbeliefs. These findings are consistent with prior research showing that personalized messages aligned with stable personality traits can increase engagement and message relevance [8,9]. The success of extraversion-targeting messages underscores the utility of leveraging personality-based characteristics in correction strategies, particularly when addressing misbeliefs in a population with diverse psychological profiles.

It is worth noting that the generic condition represented a high standard of comparison, as the messages were sourced from empirically validated resources designed to counter vaccine misinformation. The fact that the extraversion-targeting condition performed comparably, and in some cases better, suggests that tailoring messages to personality traits may enhance their salience and persuasiveness, even when compared to well-designed generic messages.

However, the effects of the extraversion-targeting messages were limited to the immediate outcome of reducing vaccine misbeliefs and did not extend to broader vaccination attitudes. This aligns with prior studies, which have found that the effects of message targeting are often confined to proximal outcomes and may not translate to more distal outcomes without sustained or multifaceted interventions [5,16]. Addressing general attitudes toward vaccination may require cumulative exposure to corrective messages or additional strategies, such as emotional appeals or peer endorsements, to reinforce the intended impact.

### 5.2. Challenges of pseudoscientific-belief-targeting messages

The pseudoscientific-belief-targeting condition, in contrast, yielded no significant reductions in vaccine misbeliefs and, in some cases, backfired by reinforcing negative vaccination attitudes among individuals with strong pseudoscientific beliefs. These findings highlight the double-edged nature of message targeting. Defensive processing and resistance to persuasion are well-documented phenomena, especially when individuals feel their beliefs are being directly challenged or manipulated [32].

Several factors may account for the ineffectiveness of pseudoscientific-belief-targeting messages. First, pseudoscientific beliefs remain a relatively new concept in the literature [11], and while the generative AI model was provided with an accurate definition, it may lack the nuanced understanding required to craft effective messages

targeting these beliefs. Additionally, although the post-hoc human validation results revealed a significant difference between the AI-generated pseudoscientific-belief-targeting messages and the generic real-world messages, this difference was relatively small compared to the difference observed for the extraversion-targeting messages. This suggests that the AI-generated messages may not have been sufficiently distinct from generic corrections to meaningfully influence belief change. One possible explanation is that the generic messages used in this study were sourced from a website dedicated to countering vaccine misinformation through empirically supported refutations. As a result, these messages may have already been designed in a way that appeals to individuals with strong pseudoscientific beliefs, making it difficult for AI to generate an even more targeted version. In previous research on message targeting, studies have often created both a low version and a high version of the targeted characteristic to allow for clear comparisons [9]. However, this study used real-world generic messages as the comparison rather than a low version to enhance the ecological validity of the study. A more extreme or distinct pseudoscientific-belief-targeting message might be needed to enhance effectiveness, but such an approach could also increase the risk of backfire effects by triggering greater resistance among individuals with entrenched pseudoscientific beliefs.

Second, belief-based targeting is likely more complex than personality-based targeting. Personality traits like extraversion are stable, broadly defined characteristics, making it easier to design flexible messages that resonate with individuals. In contrast, pseudoscientific beliefs are deeply tied to specific issues and often reinforced by cognitive biases, making them more resistant to correction. Third, directly targeting beliefs closely related to the correction context, such as pseudoscientific beliefs in the context of vaccine misinformation, may inadvertently heighten defensiveness. Misbelief correction is already a challenging task, and when individuals sense that their core beliefs are being directly targeted, they may perceive the message as manipulative or threatening. This suggests that targeting characteristics less directly tied to the correction context, such as personality traits, may be a more effective and less risky strategy for designing corrective messages.

Notably, these findings do not suggest that AI-assisted messaging as a whole is ineffective at targeting entrenched beliefs. Instead, they indicate that certain AI-generated messages may be ineffective or counterproductive, while others may be more effective depending on their construction. Given that this study tested only one version of each targeted message, the findings do not rule out the possibility that other AI-generated messages, perhaps structured differently or incorporating additional persuasive strategies, could be more effective. Future research should explore how different AI-generated message variations impact belief correction, particularly when targeting audiences with strong preexisting misbeliefs.

### 5.3. Limitations

While this study provides valuable insights, it is not without limitations. First, the correction messages used in this study were plain-text messages without visuals or additional design features. Visual elements, interactive components, or emotionally engaging content may enhance message effectiveness and should be explored in future research. Second, this study focused exclusively on two audience characteristics: extraversion and pseudoscientific beliefs. While these characteristics were chosen for their theoretical and practical relevance, there are many other characteristics, such as openness to experience, cognitive styles, or cultural values, that could influence the effectiveness of targeted messages. Expanding the scope of audience characteristics in future studies would provide a more comprehensive understanding of message targeting strategies.

Third, the sample was not nationally representative, as it included a higher proportion of women and individuals with lower household income. This sampling approach was intentional, as the study focused on

individuals with neutral or negative attitudes toward COVID-19 vaccines, a group that is both more susceptible to vaccine misinformation and more resistant to corrective interventions. However, this limits the generalizability of the findings to the broader population. Future research should explore whether the effectiveness of AI-assisted targeted correction messages holds across more demographically diverse and nationally representative samples. Examining how different population subgroups respond to these strategies would provide a clearer understanding of their applicability in real-world misinformation correction efforts.

Fourth, participants were not explicitly informed that the targeted correction messages were AI-generated. This decision was made to maintain consistency across conditions, as the control and generic messages were sourced from real-world fact-checking websites rather than generated by AI. However, this raises an important question about how transparency regarding AI involvement might influence the effectiveness of correction messages. Research suggests that disclosure of AI involvement in content creation can sometimes lead to skepticism or reduced trust, particularly in contexts where credibility is essential [33,34]. If participants had been explicitly informed that the correction messages were AI-generated, they might have responded differently, potentially discounting the message's credibility or resisting the correction. However, prior research by Matz, Teeny [8] found that AI-generated targeted messages were equally effective regardless of whether participants were informed of their origin. Importantly, Matz et al.'s [8] study did not focus on misinformation correction, so it remains unclear whether similar findings would apply in this context. Future research should examine whether transparency about AI-generated corrections influences their persuasiveness, particularly in cases where trust in the correction source plays a crucial role.

#### 5.4. Practical implications

The findings have practical implications for public health communication and misinformation correction. The success of extraversion-targeting messages demonstrates the value of incorporating personality traits into message design to enhance relevance and engagement. Public health practitioners and communicators can leverage generative AI as an assistive tool to efficiently create personalized messages for diverse audiences, making large-scale interventions more feasible and impactful.

However, the results also highlight the risks of belief-based targeting. Messages that directly challenge deeply entrenched beliefs, for instance, pseudoscientific beliefs, may trigger defensive reactions and backfire. Practitioners should exercise caution when designing messages that correct misinformation closely tied to a person's identity or worldview. Instead, targeting characteristics less directly tied to the correction context, such as personality traits, may provide a more effective approach.

Additionally, this study highlights the importance of refining AI-generated messages before deployment, rather than simply accepting the first message produced by generative AI. A more effective approach involves an iterative process, where communicators refine their prompts, generate multiple message variations, and assess their effectiveness to ensure they resonate with the target audience and do not trigger unintended resistance before dissemination. Human oversight remains essential, as small manual adjustments, such as refining language or emphasizing key persuasive elements, can significantly enhance the message's impact. Future research should explore how human-AI collaboration can be optimized to create persuasive, trustworthy, and adaptable messages for combating misinformation effectively.

## 6. Conclusion

This study demonstrates the potential of generative AI to create

effective, targeted correction messages, while also highlighting the complexities and limitations of message targeting strategies. By advancing our understanding of how audience characteristics influence the effectiveness of corrective messages, this research provides valuable insights for combating misinformation and addressing vaccine hesitancy. Future studies should build on these findings by exploring additional audience characteristics, testing diverse message formats, and examining broader populations to develop more robust and scalable solutions for misinformation correction.

## CRedit authorship contribution statement

**Hang Lu:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

## Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.vaccine.2025.127018>.

## Data availability

Data will be made available on request.

## References

- [1] Greenwood B. The contribution of vaccination to global health: past, present and future. *Philosophical Transactions of the Royal Society B: Biological Sciences* 2014; 369(1645):20130433.
- [2] World Health Organization. Ten threats to global health in 2019. Available from: <https://www.who.int/emergencies/ten-threats-to-global-health-in-2019>; 2019.
- [3] Joslyn MR, Sylvester SM. The determinants and consequences of accurate beliefs about childhood vaccinations. *Am Politics Res* 2019;47(3):628–49.
- [4] Elkin LE, Pullon SR, Stubbe MH. 'Should I vaccinate my child?' Comparing the displayed stances of vaccine information retrieved from Google. *Facebook and YouTube Vaccine* 2020;38(13):2771–8.
- [5] Whitehead HS, et al. A systematic review of communication interventions for countering vaccine misinformation. *Vaccine* 2023;41(5):1018–34.
- [6] Lunz Trujillo K, et al. Correcting misperceptions about the MMR vaccine: using psychological risk factors to inform targeted communication strategies. *Polit Res Q* 2021;74(2):464–78.
- [7] Matz S, et al. Personality science in the digital age: the promises and challenges of psychological targeting for personalized behavior-change interventions at scale. *Perspect Psychol Sci* 2024;19(6):1031–56.
- [8] Matz S, et al. The potential of generative AI for personalized persuasion at scale. *Sci Rep* 2024;14(1):4692.
- [9] Simchon A, Edwards M, Lewandowsky S. The persuasive effects of political microtargeting in the age of generative artificial intelligence. *PNAS nexus* 2024;3(2). p. pgae035.
- [10] Soto CJ, John OP. The next big five inventory (BFI-2): developing and assessing a hierarchical model with 15 facets to enhance bandwidth, fidelity, and predictive power. *J Pers Soc Psychol* 2017;113(1):117–43.
- [11] Fasce A, Avendaño D, Adrián-Ventura J. Revised and short versions of the pseudoscientific belief scale. *Appl Cogn Psychol* 2021;35(3):828–32.
- [12] Lewandowsky S, et al. Misinformation and its correction: continued influence and successful debiasing. *Psychol Sci Public Interest* 2012;13(3):106–31.
- [13] Taber CS, Lodge M. Motivated skepticism in the evaluation of political beliefs. *Am J Polit Sci* 2006;50(3):755–69.
- [14] Flynn DJ, Nyhan B, Reifler J. The nature and origins of misperceptions: understanding false and unsupported beliefs about politics. *Polit Psychol* 2017;38: 127–50.
- [15] Oliver JE, Wood TJ. *Enchanted America: How intuition & reason divide our politics*. University of Chicago Press; 2018.
- [16] Prieke T, Ecker UK. Effective correction of misinformation. *Curr Opin Psychol* 2023; 54:101712.
- [17] Moyer-Gusé E, Robinson MJ, Mcknight J. The role of humor in messaging about the MMR vaccine. *J Health Commun* 2018;23(6):514–22.
- [18] Fasce A, et al. A taxonomy of anti-vaccination arguments from a systematic literature review and text modelling. *Nat Hum Behav* 2023;7:1462–80.

- [19] Schmid KL, et al. Targeting or tailoring? Maximizing resources to create effective health communications. *Mark Health Serv* 2008;28(1):32–7.
- [20] Teeny JD, et al. A review and conceptual framework for understanding personalized matching effects in persuasion. *J Consum Psychol* 2021;31(2): 382–414.
- [21] Feinberg M, Willer R. From gulf to bridge: when do moral arguments facilitate political influence? *Personal Soc Psychol Bull* 2015;41(12):1665–81.
- [22] Chu J, Pink SL, Willer R. Religious identity cues increase vaccination intentions and trust in medical experts among American Christians. *Proc Natl Acad Sci* 2021; 118(49):e2106481118.
- [23] Holford D, et al. The empathetic refutational interview to tackle vaccine misconceptions: four randomized experiments. *Health Psychol* 2024;43(6):426–37.
- [24] Attwell K, Freeman M. I immunise: an evaluation of a values-based campaign to change attitudes and beliefs. *Vaccine* 2015;33(46):6235–40.
- [25] Simchon A, et al. Online reading habits can reveal personality traits: towards detecting psychological microtargeting. *PNAS Nexus* 2023;2(6). p. pgad191.
- [26] Matz SC, et al. Psychological targeting as an effective approach to digital mass persuasion. *Proc Natl Acad Sci* 2017;114(48):12714–9.
- [27] Costello TH, Pennycook G, Rand DG. Durably reducing conspiracy beliefs through dialogues with AI. *Science* 2024;385(6714). p. eadq1814.
- [28] Holford DL, et al. Psychological profiles of anti-vaccination argument endorsement. *Sci Rep* 2023;13:11219.
- [29] Holford D, et al. Identifying the underlying psychological constructs from self-expressed anti-vaccination argumentation. *Humanities and Social Sciences Communications* 2024;11:926.
- [30] Lu H. The role of repeated exposure and message fatigue in influencing willingness to help polar bears and support climate change mitigation. *Sci Commun* 2022;44 (4):475–93.
- [31] Hayes AF, editor. *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach*. 2nd ed. ed. New York, NY: The Guilford Press; 2017.
- [32] Brehm JW. *A theory of psychological reactance*. Social psychology. 1966. New York,: Academic Press. x, [135 p].
- [33] Chu H, Liu S. Can AI tell good stories? Narrative transportation and persuasion with ChatGPT. *J Commun* 2024;74(5):347–58.
- [34] Dai Y, Lee J, Kim JW. AI vs. human voices: how delivery source and narrative format influence the effectiveness of persuasion messages. *International Journal of Human–Computer Interaction* 2024;40(24):8735–49.